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Attention Is All You Need

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Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing both with large and limited training data.

*Equal contribution. Listing order is random. Jakob proposed replacing RNNs with self-attention and started the effort to evaluate this idea. Ashish, with Illia, designed and implemented the first Transformer models and has been crucially involved in every aspect of this work. Noam proposed scaled dot-product attention, multi-head attention and the parameter-free position representation and became the other person involved in nearly every detail. Niki designed, implemented, tuned and evaluated countless model variants in our original codebase and tensor2tensor. Llion also experimented with novel model variants, was responsible for our initial codebase, and efficient inference and visualizations. Lukasz and Aidan spent countless long days designing various parts of and implementing tensor2tensor, replacing our earlier codebase, greatly improving results and massively accelerating our research.

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摘要

主流的序列转换模型基于复杂的循环神经网络或卷积神经网络，并包含编码器和解码器。表现最佳的模型还通过注意力机制连接编码器和解码器。我们提出了一种全新的简单网络架构——Transformer，它完全基于注意力机制，彻底摒弃了循环和卷积。两项机器翻译任务的实验表明，这些模型的质量更高，同时更易于并行化，训练所需时间也显著减少。在 WMT 2014 英译德翻译任务中，我们的模型取得了 28.4 BLEU 分数，比现有最佳结果（包括集成模型）高出 2 个以上 BLEU 分数。在 WMT 2014 英译法翻译任务中，我们的模型创下了单模型 BLEU 分数的新纪录，达到 41.8；该模型在八块 GPU 上训练了 3.5 天，训练成本只是文献中最佳模型的一小部分。我们还将 Transformer 成功应用于英语成分句法分析任务，并在大规模和有限训练数据两种情况下验证了它良好的泛化能力。

*贡献相同，署名顺序随机。Jakob 提议用自注意力取代 RNN，并着手评估这一想法。Ashish 与 Illia 设计并实现了首批 Transformer 模型，并在这项工作的各个方面都发挥了关键作用。Noam 提出了缩放点积注意力、多头注意力和无参数的位置表示，成为另一位几乎参与每个细节的人。Niki 在我们的初始代码库和 tensor2tensor 中设计、实现、调整并评估了无数种模型变体。Llion 也尝试了新的模型变体，负责搭建我们的初始代码库，并实现了高效推理和可视化。Lukasz 和 Aidan 花了无数个漫长的日夜，设计并实现 tensor2tensor 的各个部分，以取代我们早期的代码库，大幅提升结果，并极大地加速了我们的研究。

[†]在 Google Brain 工作期间完成。

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1 Introduction

Recurrent neural networks, long short-term memory [13] and gated recurrent [7] neural networks in particular, have been firmly established as state of the art approaches in sequence modeling and transduction problems such as language modeling and machine translation [35, 2, 5]. Numerous efforts have since continued to push the boundaries of recurrent language models and encoder-decoder architectures [38, 24, 15].

Recurrent models typically factor computation along the symbol positions of the input and output sequences. Aligning the positions to steps in computation time, they generate a sequence of hidden states h_t , as a function of the previous hidden state h_{t-1} and the input for position t . This inherently sequential nature precludes parallelization within training examples, which becomes critical at longer sequence lengths, as memory constraints limit batching across examples. Recent work has achieved significant improvements in computational efficiency through factorization tricks [21] and conditional computation [32], while also improving model performance in case of the latter. The fundamental constraint of sequential computation, however, remains.

Attention mechanisms have become an integral part of compelling sequence modeling and transduction models in various tasks, allowing modeling of dependencies without regard to their distance in the input or output sequences [2, 19]. In all but a few cases [27], however, such attention mechanisms are used in conjunction with a recurrent network.

In this work we propose the Transformer, a model architecture eschewing recurrence and instead relying entirely on an attention mechanism to draw global dependencies between input and output. The Transformer allows for significantly more parallelization and can reach a new state of the art in translation quality after being trained for as little as twelve hours on eight P100 GPUs.

2 Background

The goal of reducing sequential computation also forms the foundation of the Extended Neural GPU [16], ByteNet [18] and ConvS2S [9], all of which use convolutional neural networks as basic building block, computing hidden representations in parallel for all input and output positions. In these models, the number of operations required to relate signals from two arbitrary input or output positions grows in the distance between positions, linearly for ConvS2S and logarithmically for ByteNet. This makes it more difficult to learn dependencies between distant positions [12]. In the Transformer this is reduced to a constant number of operations, albeit at the cost of reduced effective resolution due to averaging attention-weighted positions, an effect we counteract with Multi-Head Attention as described in section 3.2.

Self-attention, sometimes called intra-attention is an attention mechanism relating different positions of a single sequence in order to compute a representation of the sequence. Self-attention has been used successfully in a variety of tasks including reading comprehension, abstractive summarization, textual entailment and learning task-independent sentence representations [4, 27, 28, 22].

End-to-end memory networks are based on a recurrent attention mechanism instead of sequence-aligned recurrence and have been shown to perform well on simple-language question answering and language modeling tasks [34].

To the best of our knowledge, however, the Transformer is the first transduction model relying entirely on self-attention to compute representations of its input and output without using sequence-aligned RNNs or convolution. In the following sections, we will describe the Transformer, motivate self-attention and discuss its advantages over models such as [17, 18] and [9].

3 Model Architecture

Most competitive neural sequence transduction models have an encoder-decoder structure [5, 2, 35]. Here, the encoder maps an input sequence of symbol representations $(x_1, \dots, x_n)$ to a sequence of continuous representations $\mathbf{z} = (z_1, \dots, z_n)$. Given $\mathbf{z}$, the decoder then generates an output sequence $(y_1, \dots, y_m)$ of symbols one element at a time. At each step the model is auto-regressive [10], consuming the previously generated symbols as additional input when generating the next.

1 引言

循环神经网络，尤其是长短期记忆网络 [13] 和门控循环 [7] 神经网络，已成为序列建模和转换问题（如语言建模和机器翻译）中的最先进方法 [35, 2, 5]。此后，众多研究持续推动循环语言模型和编码器—解码器架构的发展 [38, 24, 15]。

循环模型通常沿着输入和输出序列的符号位置分解计算。将这些位置与计算时间步对齐后，模型会根据前一个隐藏状态 h_{t-1} 和位置 t 的输入，生成一系列隐藏状态 h_t 。这种固有的顺序性使得训练样本内部无法并行计算；当序列较长时，这一点尤为关键，因为内存限制会制约跨样本的批处理。近期研究通过因式分解技巧 [21]和条件计算 [32], , 显著提高了计算效率；后者还提升了模型性能。然而，顺序计算这一根本限制依然存在。

注意力机制已成为各类任务中高性能序列建模和转导模型不可或缺的一部分，使模型能够不受输入或输出序列中依赖距离的限制来建模依赖关系 [2, 19]。然而，除少数情况 [27], 外，这类注意力机制通常与循环网络结合使用。

在这项工作中，我们提出了 Transformer：一种摒弃循环结构、完全依靠注意力机制来建立输入与输出之间全局依赖关系的模型架构。Transformer 的并行能力显著更强，仅需在 8 块 P100 GPU 上训练 12 小时，就能在翻译质量上达到新的最先进水平。

2 背景

减少顺序计算的目标也是 Extended Neural GPU[16]、ByteNet [18] 和 ConvS2S [9] 的基础。这些模型都使用卷积神经网络作为基本构建模块，并行计算所有输入和输出位置的隐藏表示。在这些模型中，关联任意两个输入或输出位置的信号所需的操作次数会随着位置间距离的增加而增加：对于 ConvS2S，这种增长是线性的；对于 ByteNet，则是对数级的。这使得模型更难学习相距较远位置之间的依赖关系 [12]。在 Transformer 中，这一操作次数减少为常数，但代价是由于对注意力加权的位置进行平均，有效分辨率有所降低。正如第 3.2 节所述，我们通过多头注意力来抵消这一影响。

自注意力有时也称为内部注意力，是一种关联单个序列中不同位置的注意力机制，用于计算该序列的表示。自注意力已成功应用于多种任务，包括阅读理解、抽象式摘要、文本蕴含以及及与任务无关的句子表示学习 [4, 27, 28, 22]。

端到端记忆网络基于循环注意力机制，而非序列对齐的循环结构，并已证明在简单语言问答和语言建模任务上表现良好 [34]。

据我们所知，然而，Transformer 是首个完全依赖自注意力来计算输入和输出表示、而不使用序列对齐 RNN 或卷积的转导模型。在接下来的章节中，我们将介绍 Transformer，阐述自注意力的动机，并讨论它相较于 [17, 18] 和 [9]等模型的优势。

3 模型架构

大多数具有竞争力的神经序列转导模型都采用编码器-解码器结构 [5, 2, 35]。在这里，编码器将输入符号表示序列 $(x_1, \dots, x_n)$ 映射为连续表示序列 $\mathbf{z} = (z_1, \dots, z_n)$ 。给定 $\mathbf{z}$ 后，解码器会逐个生成输出符号序列 $(y_1, \dots, y_m)$ 。在每一步中，模型都以自回归方式运行[10], , 在生成下一个符号时，将此前生成的符号作为额外输入。

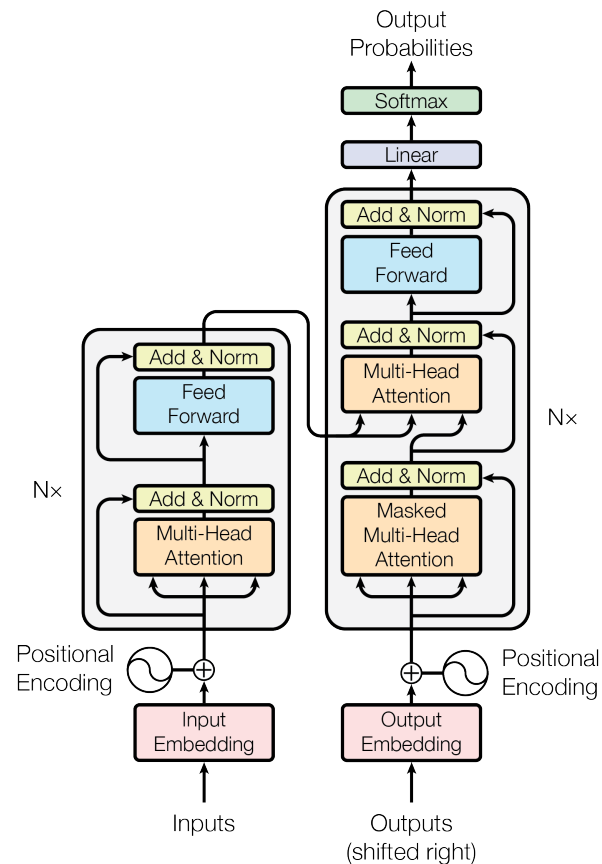


Figure 1: The Transformer - model architecture.

The Transformer follows this overall architecture using stacked self-attention and point-wise, fully connected layers for both the encoder and decoder, shown in the left and right halves of Figure 1, respectively.

3.1 Encoder and Decoder Stacks

Encoder: The encoder is composed of a stack of $N = 6$ identical layers. Each layer has two sub-layers. The first is a multi-head self-attention mechanism, and the second is a simple, position-wise fully connected feed-forward network. We employ a residual connection [11] around each of the two sub-layers, followed by layer normalization [1]. That is, the output of each sub-layer is $\text{LayerNorm}(x + \text{Sublayer}(x))$, where $\text{Sublayer}(x)$ is the function implemented by the sub-layer itself. To facilitate these residual connections, all sub-layers in the model, as well as the embedding layers, produce outputs of dimension $d_{\text{model}} = 512$.

Decoder: The decoder is also composed of a stack of $N = 6$ identical layers. In addition to the two sub-layers in each encoder layer, the decoder inserts a third sub-layer, which performs multi-head attention over the output of the encoder stack. Similar to the encoder, we employ residual connections around each of the sub-layers, followed by layer normalization. We also modify the self-attention sub-layer in the decoder stack to prevent positions from attending to subsequent positions. This masking, combined with fact that the output embeddings are offset by one position, ensures that the predictions for position i can depend only on the known outputs at positions less than i .

3.2 Attention

An attention function can be described as mapping a query and a set of key-value pairs to an output, where the query, keys, values, and output are all vectors. The output is computed as a weighted sum

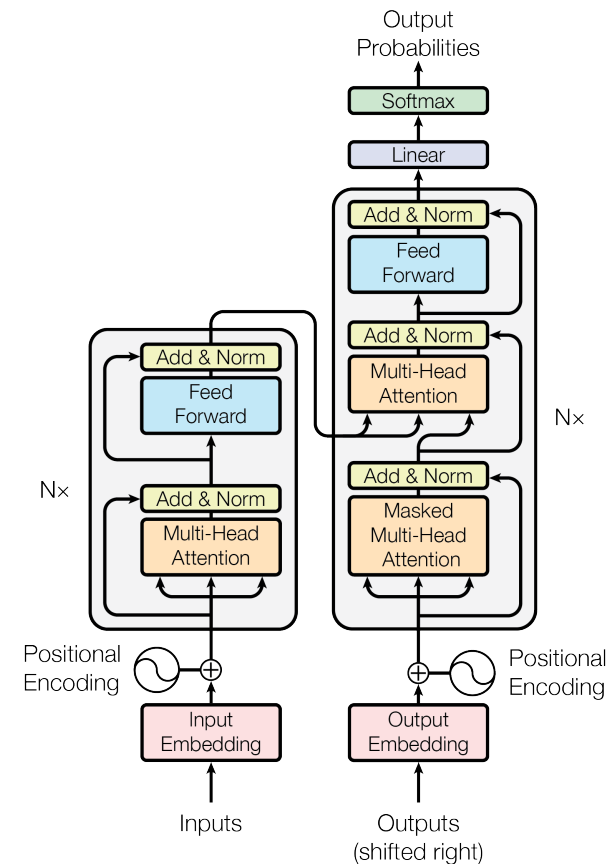


图 1: Transformer 模型架构。

Transformer 遵循这一整体架构，编码器和解码器均采用堆叠的自注意力层和逐点全连接层，如图 1 左右两部分所示。

3.1 编码器和解码器堆栈

编码器：编码器由 $N = 6$ 个相同的层堆叠而成。每层包含两个子层。第一个是多头自注意力机制，第二个是简单的、逐位置的全连接前馈网络。我们在这两个子层周围分别采用残差连接 [11]，然后进行层归一化 [1]。也就是说，每个子层的输出为 $\text{LayerNorm}(x + \text{Sublayer}(x))$ ，其中 $\text{Sublayer}(x)$ 是子层本身实现的函数。为便于使用这些残差连接，模型中的所有子层以及嵌入层的输出维度均为 $d_{\text{model}} = 512$ 。

解码器：解码器同样由 $N = 6$ 个相同的层堆叠而成。除了每个编码器层中的两个子层外，解码器还插入了第三个子层，用于对编码器堆栈的输出执行多头注意力。与编码器类似，我们在每个子层周围采用残差连接，然后进行层归一化。我们还修改了解码器堆栈中的自注意力子层，以防止当前位置关注后续位置。这种掩码与输出嵌入向前偏移一个位置相结合，确保位置 i 的预测只能依赖于位置小于 i 的已知输出。

3.2 注意力

注意力函数可以描述为将一个查询和一组键值对映射为一个输出，其中查询、键、值和输出都是向量。输出是一个加权和



Figure 2: (left) Scaled Dot-Product Attention. (right) Multi-Head Attention consists of several attention layers running in parallel.

of the values, where the weight assigned to each value is computed by a compatibility function of the query with the corresponding key.

3.2.1 Scaled Dot-Product Attention

We call our particular attention "Scaled Dot-Product Attention" (Figure 2). The input consists of queries and keys of dimension d_k , and values of dimension d_v . We compute the dot products of the query with all keys, divide each by $\sqrt{d_k}$, and apply a softmax function to obtain the weights on the values.

In practice, we compute the attention function on a set of queries simultaneously, packed together into a matrix Q . The keys and values are also packed together into matrices K and V . We compute the matrix of outputs as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

The two most commonly used attention functions are additive attention [2], and dot-product (multiplicative) attention. Dot-product attention is identical to our algorithm, except for the scaling factor of $\frac{1}{\sqrt{d_k}}$. Additive attention computes the compatibility function using a feed-forward network with a single hidden layer. While the two are similar in theoretical complexity, dot-product attention is much faster and more space-efficient in practice, since it can be implemented using highly optimized matrix multiplication code.

While for small values of d_k the two mechanisms perform similarly, additive attention outperforms dot product attention without scaling for larger values of d_k [3]. We suspect that for large values of d_k , the dot products grow large in magnitude, pushing the softmax function into regions where it has extremely small gradients⁴. To counteract this effect, we scale the dot products by $\frac{1}{\sqrt{d_k}}$.

3.2.2 Multi-Head Attention

Instead of performing a single attention function with d_{model} -dimensional keys, values and queries, we found it beneficial to linearly project the queries, keys and values h times with different, learned linear projections to d_k , d_k and d_v dimensions, respectively. On each of these projected versions of queries, keys and values we then perform the attention function in parallel, yielding d_v -dimensional

⁴To illustrate why the dot products get large, assume that the components of q and k are independent random variables with mean 0 and variance 1. Then their dot product, $q \cdot k = \sum_{i=1}^{d_k} q_i k_i$, has mean 0 and variance d_k .

图 2：（左）缩放点积注意力。（右）多头注意力由多个并行运行的注意力层组成。

由各个值构成，其中，每个值所分配的权重由一个兼容性函数根据查询与相应键之间的匹配程度计算得出。

3.2.1 缩放点积注意力

我们将所采用的特定注意力机制称为“缩放点积注意力”（图 2）。输入包括维度为 d_k 的查询和键，以及维度为 d_v 的值。我们计算查询与所有键的点积，将每个点积除以 $\sqrt{d_k}$ ，然后应用 softmax 函数，得到各个值对应的权重。

在实践中，我们会将一组查询打包成矩阵 Q ，并同时计算它们的注意力函数。键和值也分别打包成矩阵 K 和 V 。输出矩阵的计算方式如下：

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

最常用的两种注意力函数是加性注意力 [2]，和点积（乘法）注意力。点积注意力与我们的算法相同，区别仅在于缩放因子 $\frac{1}{\sqrt{d_k}}$ 。加性注意力使用只有一个隐藏层的前馈网络来计算兼容性函数。虽然这两种方法在理论复杂度上相近，但在实际应用中，点积注意力速度快得多，也更节省空间，因为它可以通过高度优化的矩阵乘法代码实现。

当 d_k 的值较小时，这两种机制的表现相近；但当 d_k [3]较大时，加性注意力的表现优于未经缩放的点积注意力。我们推测，当 d_k 较大时，点积的数值会变得很大，使 softmax 函数进入梯度极小的区域⁴。为抵消这一影响，我们将点积按 $\frac{1}{\sqrt{d_k}}$ 进行缩放。

3.2.2 多头注意力

我们发现，与其使用 d_{model} 维的键、值和查询执行单个注意力函数，不如将查询、键和值分别通过 h 组不同的可学习线性投影，映射到 d_k 、 d_k 和 d_v 维空间。然后，我们在这些投影后的查询、键和值上并行执行注意力函数，得到 d_v 维的

⁴为说明点积为何会变得很大，假设 q 和 k 的各个分量是均值为 0、方差为 1 的独立随机变量。那么，它们的点积 $q \cdot k = \sum_{i=1}^{d_k} q_i k_i$ 的均值为 0，方差为 d_k 。

output values. These are concatenated and once again projected, resulting in the final values, as depicted in Figure 2.

Multi-head attention allows the model to jointly attend to information from different representation subspaces at different positions. With a single attention head, averaging inhibits this.

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O$$

$$\text{where head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$$

Where the projections are parameter matrices $W_i^Q \in \mathbb{R}^{d_{\text{model}} \times d_k}$, $W_i^K \in \mathbb{R}^{d_{\text{model}} \times d_k}$, $W_i^V \in \mathbb{R}^{d_{\text{model}} \times d_v}$ and $W^O \in \mathbb{R}^{hd_v \times d_{\text{model}}}$.

In this work we employ $h = 8$ parallel attention layers, or heads. For each of these we use $d_k = d_v = d_{\text{model}}/h = 64$. Due to the reduced dimension of each head, the total computational cost is similar to that of single-head attention with full dimensionality.

3.2.3 Applications of Attention in our Model

The Transformer uses multi-head attention in three different ways:

- In "encoder-decoder attention" layers, the queries come from the previous decoder layer, and the memory keys and values come from the output of the encoder. This allows every position in the decoder to attend over all positions in the input sequence. This mimics the typical encoder-decoder attention mechanisms in sequence-to-sequence models such as [38, 2, 9].
- The encoder contains self-attention layers. In a self-attention layer all of the keys, values and queries come from the same place, in this case, the output of the previous layer in the encoder. Each position in the encoder can attend to all positions in the previous layer of the encoder.
- Similarly, self-attention layers in the decoder allow each position in the decoder to attend to all positions in the decoder up to and including that position. We need to prevent leftward information flow in the decoder to preserve the auto-regressive property. We implement this inside of scaled dot-product attention by masking out (setting to $-\infty$) all values in the input of the softmax which correspond to illegal connections. See Figure 2.

3.3 Position-wise Feed-Forward Networks

In addition to attention sub-layers, each of the layers in our encoder and decoder contains a fully connected feed-forward network, which is applied to each position separately and identically. This consists of two linear transformations with a ReLU activation in between.

$$\text{FFN}(x) = \max(0, xW_1 + b_1)W_2 + b_2 \quad (2)$$

While the linear transformations are the same across different positions, they use different parameters from layer to layer. Another way of describing this is as two convolutions with kernel size 1. The dimensionality of input and output is $d_{\text{model}} = 512$, and the inner-layer has dimensionality $d_{ff} = 2048$.

3.4 Embeddings and Softmax

Similarly to other sequence transduction models, we use learned embeddings to convert the input tokens and output tokens to vectors of dimension d_{model} . We also use the usual learned linear transformation and softmax function to convert the decoder output to predicted next-token probabilities. In our model, we share the same weight matrix between the two embedding layers and the pre-softmax linear transformation, similar to [30]. In the embedding layers, we multiply those weights by $\sqrt{d_{\text{model}}}$.

输出值。这些输出会被拼接起来，再次进行投影，从而得到最终值，如图 2 所示。

多头注意力使模型能够在不同位置联合关注来自不同表示子空间的信息。使用单个注意力头时，取平均会妨碍这一点。

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O$$

$$\text{where head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$$

其中，投影矩阵为参数矩阵 $W_i^Q \in \mathbb{R}^{d_{\text{model}} \times d_k}$ 、 $W_i^K \in \mathbb{R}^{d_{\text{model}} \times d_k}$ 、 $W_i^V \in \mathbb{R}^{d_{\text{model}} \times d_v}$ 和 $W^O \in \mathbb{R}^{hd_v \times d_{\text{model}}}$ 。

本文采用 $h = 8$ 个并行注意力层，即注意力头。对于每个注意力头，我们使用 $d_k = d_v = d_{\text{model}}/h = 64$ 。由于每个头的维度较低，总计算成本与全维度的单头注意力相近。

3.2.3 我们模型中的注意力应用

Transformer 以三种不同方式使用多头注意力：

- 在“编码器-解码器注意力”层中，查询来自前一个解码器层，而记忆键和值则来自编码器的输出。这使解码器中的每个位置都能关注输入序列中的所有位置。这模拟了序列到序列模型中典型的编码器-解码器注意力机制，例如[38, 2, 9]。
- 编码器包含自注意力层。在自注意力层中，所有键、值和查询都来自同一位置，在本例中即编码器前一层的输出。编码器中的每个位置都可以关注编码器前一层中的所有位置。
- 类似地，解码器中的自注意力层允许解码器中的每个位置关注解码器中截至该位置（包括该位置）的所有位置。为了保留自回归特性，我们需要阻止信息在解码器中向左流动。我们在缩放点积注意力中通过屏蔽所有对应非法连接的 softmax 输入值（将其设为 $-\infty$ ）来实现这一点。参见图 2。

3.3 逐位置前馈网络

除了注意力子层之外，我们的编码器和解码器中的每一层都包含一个全连接前馈网络，该网络以相同方式独立应用于每个位置。它由两个线性变换组成，中间使用 ReLU 激活函数。

$$\text{FFN}(x) = \max(0, xW_1 + b_1)W_2 + b_2 \quad (2)$$

虽然线性变换在不同位置上相同，但不同层使用的参数不同。另一种描述方式是将其视为两个卷积，卷积核大小为 1。输入和输出的维度为 $d_{\text{model}} = 512$ ，内层的维度为 $d_{ff} = 2048$ 。

3.4 嵌入与 Softmax

与其他序列转换模型类似，我们使用学习得到的嵌入，将输入词元和输出词元转换为维度为 d_{model} 的向量。我们还使用常见的学习得到的线性变换和 softmax 函数，将解码器的输出转换为预测的下一个词元概率。在我们的模型中，两个嵌入层与 softmax 之前的线性变换共享同一个权重矩阵，这与 [30] 类似。在嵌入层中，我们将这些权重乘以 $\sqrt{d_{\text{model}}}$ 。

Table 1: Maximum path lengths, per-layer complexity and minimum number of sequential operations for different layer types. n is the sequence length, d is the representation dimension, k is the kernel size of convolutions and r the size of the neighborhood in restricted self-attention.

Layer Type	Complexity per Layer	Sequential Operations	Maximum Path Length
Self-Attention	$O(n^2 \cdot d)$	$O(1)$	$O(1)$
Recurrent	$O(n \cdot d^2)$	$O(n)$	$O(n)$
Convolutional	$O(k \cdot n \cdot d^2)$	$O(1)$	$O(\log_k(n))$
Self-Attention (restricted)	$O(r \cdot n \cdot d)$	$O(1)$	$O(n/r)$

3.5 Positional Encoding

Since our model contains no recurrence and no convolution, in order for the model to make use of the order of the sequence, we must inject some information about the relative or absolute position of the tokens in the sequence. To this end, we add "positional encodings" to the input embeddings at the bottoms of the encoder and decoder stacks. The positional encodings have the same dimension d_{model} as the embeddings, so that the two can be summed. There are many choices of positional encodings, learned and fixed [9].

In this work, we use sine and cosine functions of different frequencies:

$$PE_{(pos, 2i)} = \sin(pos/10000^{2i/d_{\text{model}}})$$

$$PE_{(pos, 2i+1)} = \cos(pos/10000^{2i/d_{\text{model}}})$$

where pos is the position and i is the dimension. That is, each dimension of the positional encoding corresponds to a sinusoid. The wavelengths form a geometric progression from 2π to $10000 \cdot 2\pi$. We chose this function because we hypothesized it would allow the model to easily learn to attend by relative positions, since for any fixed offset k , PE_{pos+k} can be represented as a linear function of PE_{pos} .

We also experimented with using learned positional embeddings [9] instead, and found that the two versions produced nearly identical results (see Table 3 row (E)). We chose the sinusoidal version because it may allow the model to extrapolate to sequence lengths longer than the ones encountered during training.

4 Why Self-Attention

In this section we compare various aspects of self-attention layers to the recurrent and convolutional layers commonly used for mapping one variable-length sequence of symbol representations $(x_1, \dots, x_n)$ to another sequence of equal length $(z_1, \dots, z_n)$, with $x_i, z_i \in \mathbb{R}^d$, such as a hidden layer in a typical sequence transduction encoder or decoder. Motivating our use of self-attention we consider three desiderata.

One is the total computational complexity per layer. Another is the amount of computation that can be parallelized, as measured by the minimum number of sequential operations required.

The third is the path length between long-range dependencies in the network. Learning long-range dependencies is a key challenge in many sequence transduction tasks. One key factor affecting the ability to learn such dependencies is the length of the paths forward and backward signals have to traverse in the network. The shorter these paths between any combination of positions in the input and output sequences, the easier it is to learn long-range dependencies [12]. Hence we also compare the maximum path length between any two input and output positions in networks composed of the different layer types.

As noted in Table 1, a self-attention layer connects all positions with a constant number of sequentially executed operations, whereas a recurrent layer requires $O(n)$ sequential operations. In terms of computational complexity, self-attention layers are faster than recurrent layers when the sequence

表 1: 不同层类型的最大路径长度、逐层复杂度和最少顺序操作数。 n 是序列长度, d 是表示维度, k 是卷积核大小, r 是受限自注意力中的邻域大小。

层类型	逐层复杂度	顺序操作	最大路径长度
自注意力	$O(n^2 \cdot d)$	$O(1)$	$O(1)$
循环	$O(n \cdot d^2)$	$O(n)$	$O(n)$
卷积	$O(k \cdot n \cdot d^2)$	$O(1)$	$O(\log_k(n))$
自注意力 (受限)	$O(r \cdot n \cdot d)$	$O(1)$	$O(n/r)$

3.5 位置编码

由于我们的模型不包含循环结构, 也不包含卷积结构, 为了让模型利用序列的顺序信息, 我们必须注入一些关于序列中词元相对位置或绝对位置的信息。为此, 我们在编码器和解码器堆栈底部的输入嵌入中加入“位置编码”。位置编码与嵌入的维度相同, 即 d_{model} , 因此二者可以相加。位置编码有许多选择, 包括学习得到的和固定的 [9]。

在本文中, 我们使用不同频率的正弦和余弦函数:

$$PE_{(pos, 2i)} = \sin(pos/10000^{2i/d_{\text{model}}})$$

$$PE_{(pos, 2i+1)} = \cos(pos/10000^{2i/d_{\text{model}}})$$

其中 pos 表示位置, i 表示维度。也就是说, 位置编码的每个维度都对应一个正弦函数。波长从 2π 到 $10000 \cdot 2\pi$ 按几何级数递增。我们选择这个函数, 是因为我们假设它能让模型轻松学会关注相对位置: 对于任意固定偏移量 k , PE_{pos+k} 都可以表示为 PE_{pos} 的线性函数。

我们还尝试改用学习得到的位置嵌入 [9], 发现这两种版本产生的结果几乎相同 (见表 3 第 (E) 行)。我们选择正弦函数版本, 是因为它可能使模型能够外推到比训练期间遇到的序列长度更长的序列。

4 为什么选择自注意力

本节将自注意力层的各个方面与常用于将一个可变长度的符号表示序列 $(x_1, \dots, x_n)$ 映射为另一个等长序列 $(z_1, \dots, z_n)$ 的循环层和卷积层进行比较, 其中 $x_i, z_i \in \mathbb{R}^d$, 例如典型序列转换编码器或解码器中的隐藏层。为了说明我们采用自注意力的动机, 我们考虑三个设计目标。

一项是每层的总计算复杂度。另一项是可并行计算的工作量, 以所需的最少顺序操作数来衡量。

第三项是网络中长距离依赖关系的路径长度。学习长距离依赖关系是许多序列转换任务中的一项关键挑战。影响学习这类依赖关系能力的一个关键因素, 是前向和反向信号必须在网络中经过的路径长度。输入序列和输出序列中任意位置组合之间的路径越短, 就越容易学习长距离依赖关系 [12]。因此, 我们还比较了由不同类型层组成的网络中, 任意两个输入位置和输出位置之间的最大路径长度。

如表 1 所示, 自注意力层通过固定数量的顺序执行操作连接所有位置, 而循环层则需要 $O(n)$ 个顺序操作。从计算复杂度来看, 当序列

length n is smaller than the representation dimensionality d , which is most often the case with sentence representations used by state-of-the-art models in machine translations, such as word-piece [38] and byte-pair [31] representations. To improve computational performance for tasks involving very long sequences, self-attention could be restricted to considering only a neighborhood of size r in the input sequence centered around the respective output position. This would increase the maximum path length to $O(n/r)$. We plan to investigate this approach further in future work.

A single convolutional layer with kernel width $k < n$ does not connect all pairs of input and output positions. Doing so requires a stack of $O(n/k)$ convolutional layers in the case of contiguous kernels, or $O(\log_k(n))$ in the case of dilated convolutions [18], increasing the length of the longest paths between any two positions in the network. Convolutional layers are generally more expensive than recurrent layers, by a factor of k . Separable convolutions [6], however, decrease the complexity considerably, to $O(k \cdot n \cdot d + n \cdot d^2)$. Even with $k = n$, however, the complexity of a separable convolution is equal to the combination of a self-attention layer and a point-wise feed-forward layer, the approach we take in our model.

As side benefit, self-attention could yield more interpretable models. We inspect attention distributions from our models and present and discuss examples in the appendix. Not only do individual attention heads clearly learn to perform different tasks, many appear to exhibit behavior related to the syntactic and semantic structure of the sentences.

5 Training

This section describes the training regime for our models.

5.1 Training Data and Batching

We trained on the standard WMT 2014 English-German dataset consisting of about 4.5 million sentence pairs. Sentences were encoded using byte-pair encoding [3], which has a shared source-target vocabulary of about 37000 tokens. For English-French, we used the significantly larger WMT 2014 English-French dataset consisting of 36M sentences and split tokens into a 32000 word-piece vocabulary [38]. Sentence pairs were batched together by approximate sequence length. Each training batch contained a set of sentence pairs containing approximately 25000 source tokens and 25000 target tokens.

5.2 Hardware and Schedule

We trained our models on one machine with 8 NVIDIA P100 GPUs. For our base models using the hyperparameters described throughout the paper, each training step took about 0.4 seconds. We trained the base models for a total of 100,000 steps or 12 hours. For our big models,(described on the bottom line of table 3), step time was 1.0 seconds. The big models were trained for 300,000 steps (3.5 days).

5.3 Optimizer

We used the Adam optimizer [20] with $\beta_1 = 0.9$, $\beta_2 = 0.98$ and $\epsilon = 10^{-9}$. We varied the learning rate over the course of training, according to the formula:

$$lrate = d_{\text{model}}^{-0.5} \cdot \min(step_num^{-0.5}, step_num \cdot warmup_steps^{-1.5}) \quad (3)$$

This corresponds to increasing the learning rate linearly for the first $warmup_steps$ training steps, and decreasing it thereafter proportionally to the inverse square root of the step number. We used $warmup_steps = 4000$.

5.4 Regularization

We employ three types of regularization during training:

长度 n 小于表示维度 d 时, 自注意力层比循环层更快; 对于机器翻译领域最先进模型所使用的句子表示而言, 通常就是这种情况, 例如词片段[38] 和字节对 [31] 表示。为了提升涉及超长序列的任务的计算性能, 可以将自注意力限制为仅考虑输入序列中以相应输出位置为中心、大小为 r 的邻域。这样会将最大路径长度增加到 $O(n/r)$ 。我们计划在未来的工作中进一步研究这一方法。

单个卷积层的核宽度为 $k < n$ 时, 并不能连接所有输入位置与输出位置。要实现这种连接, 在使用连续卷积核时需要堆叠 $O(n/k)$ 个卷积层; 在使用膨胀卷积时则需要堆叠 $O(\log_k(n))$ 个卷积层 [18], , 这会增加网络中任意两个位置之间最长路径的长度。卷积层通常比循环层的计算开销更大, 大约高出 k 倍。不过, 深度可分离卷积 [6], 可显著降低复杂度, 降至 $O(k \cdot n \cdot d + n \cdot d^2)$ 。然而, 即使采用 $k = n$, 深度可分离卷积的复杂度也与自注意力层和逐点前馈层的组合相当, 而这正是我们模型采用的方法。

此外, 自注意力还可能带来更具可解释性的模型。我们检查了模型中的注意力分布, 并在附录中展示和讨论了一些示例。不仅不同的注意力头显然学会了执行不同的任务, 其中许多似乎还表现出与句子的句法和语义结构相关的行为。

5 训练

本节介绍我们模型的训练方案。

5.1 训练数据与批处理

我们使用标准的 WMT 2014 英德数据集进行训练, 该数据集包含约 450 万个句对。句子采用字节对编码 [3], , 其源语言和目标语言共享约 37000 个词元的词汇表。对于英法翻译, 我们使用了规模大得多的 WMT 2014 英法数据集, 其中包含 3600 万个句子, 并将词元划分为一个包含 32000 个词片的词汇表 [38]。我们根据近似序列长度将句对组成批次。每个训练批次包含一组句对, 其中源语言词元约 25000 个, 目标语言词元也约 25000 个。

5.2 硬件与训练计划

我们使用一台配备 8 块 NVIDIA P100 GPU 的机器训练模型。对于采用本文所述超参数的基础模型, 每个训练步骤耗时约 0.4 秒。基础模型总共训练了 100,000 步, 耗时 12 小时。对于大型模型(见表 3 最后一行), 每步耗时 1.0 秒。大型模型训练了 300,000 步(耗时 3.5 天)。

5.3 优化器

我们使用 Adam 优化器 [20], 其中 $\beta_1 = 0.9$ 、 $\beta_2 = 0.98$ 和 $\epsilon = 10^{-9}$ 。在训练过程中, 我们根据以下公式调整学习率:

$$lrate = d_{\text{model}}^{-0.5} \cdot \min(step_num^{-0.5}, step_num \cdot warmup_steps^{-1.5}) \quad (3)$$

这相当于在最初的 $warmup_steps$ 个训练步骤中线性提高学习率, 之后则按照步数平方根的倒数成比例地降低学习率。我们使用 $warmup_steps = 4000$ 。

5.4 正则化

训练过程中, 我们采用三种正则化方法:

Table 2: The Transformer achieves better BLEU scores than previous state-of-the-art models on the English-to-German and English-to-French newstest2014 tests at a fraction of the training cost.

Model	BLEU		Training Cost (FLOPs)	
	EN-DE	EN-FR	EN-DE	EN-FR
ByteNet [18]	23.75			
Deep-Att + PosUnk [39]		39.2		$1.0 \cdot 10^{20}$
GNMT + RL [38]	24.6	39.92	$2.3 \cdot 10^{19}$	$1.4 \cdot 10^{20}$
ConvS2S [9]	25.16	40.46	$9.6 \cdot 10^{18}$	$1.5 \cdot 10^{20}$
MoE [32]	26.03	40.56	$2.0 \cdot 10^{19}$	$1.2 \cdot 10^{20}$
Deep-Att + PosUnk Ensemble [39]		40.4		$8.0 \cdot 10^{20}$
GNMT + RL Ensemble [38]	26.30	41.16	$1.8 \cdot 10^{20}$	$1.1 \cdot 10^{21}$
ConvS2S Ensemble [9]	26.36	41.29	$7.7 \cdot 10^{19}$	$1.2 \cdot 10^{21}$
Transformer (base model)	27.3	38.1	$3.3 \cdot 10^{18}$	
Transformer (big)	28.4	41.8	$2.3 \cdot 10^{19}$	

Residual Dropout We apply dropout [33] to the output of each sub-layer, before it is added to the sub-layer input and normalized. In addition, we apply dropout to the sums of the embeddings and the positional encodings in both the encoder and decoder stacks. For the base model, we use a rate of $P_{drop} = 0.1$.

Label Smoothing During training, we employed label smoothing of value $\epsilon_{ls} = 0.1$ [36]. This hurts perplexity, as the model learns to be more unsure, but improves accuracy and BLEU score.

6 Results

6.1 Machine Translation

On the WMT 2014 English-to-German translation task, the big transformer model (Transformer (big) in Table 2) outperforms the best previously reported models (including ensembles) by more than 2.0 BLEU, establishing a new state-of-the-art BLEU score of 28.4. The configuration of this model is listed in the bottom line of Table 3. Training took 3.5 days on 8 P100 GPUs. Even our base model surpasses all previously published models and ensembles, at a fraction of the training cost of any of the competitive models.

On the WMT 2014 English-to-French translation task, our big model achieves a BLEU score of 41.0, outperforming all of the previously published single models, at less than 1/4 the training cost of the previous state-of-the-art model. The Transformer (big) model trained for English-to-French used dropout rate $P_{drop} = 0.1$, instead of 0.3.

For the base models, we used a single model obtained by averaging the last 5 checkpoints, which were written at 10-minute intervals. For the big models, we averaged the last 20 checkpoints. We used beam search with a beam size of 4 and length penalty $\alpha = 0.6$ [38]. These hyperparameters were chosen after experimentation on the development set. We set the maximum output length during inference to input length + 50, but terminate early when possible [38].

Table 2 summarizes our results and compares our translation quality and training costs to other model architectures from the literature. We estimate the number of floating point operations used to train a model by multiplying the training time, the number of GPUs used, and an estimate of the sustained single-precision floating-point capacity of each GPU ⁵.

6.2 Model Variations

To evaluate the importance of different components of the Transformer, we varied our base model in different ways, measuring the change in performance on English-to-German translation on the

⁵We used values of 2.8, 3.7, 6.0 and 9.5 TFLOPS for K80, K40, M40 and P100, respectively.

表 2: Transformer 以极低的训练成本，在英德和英法 newstest2014 测试集上的 BLEU 得分均优于此前最先进的模型。

模型	BLEU		训练成本 (FLOPs)	
	英德	英法	英德	英法
ByteNet [18]	23.75			
Deep-Att + PosUnk [39]		39.2		$1.0 \cdot 10^{20}$
GNMT +RL [38]	24.6	39.92	$2.3 \cdot 10^{19}$	$1.4 \cdot 10^{20}$
ConvS2S [9]	25.16	40.46	$9.6 \cdot 10^{18}$	$1.5 \cdot 10^{20}$
MoE [32]	26.03	40.56	$2.0 \cdot 10^{19}$	$1.2 \cdot 10^{20}$
Deep-Att +PosUnk 集成 [39]		40.4		$8.0 \cdot 10^{20}$
GNMT +RL 集成 [38]	26.30	41.16	$1.8 \cdot 10^{20}$	$1.1 \cdot 10^{21}$
ConvS2S 集成 [9]	26.36	41.29	$7.7 \cdot 10^{19}$	$1.2 \cdot 10^{21}$
Transformer (基础模型)	27.3	38.1	$3. \cdot 10^{18}$	
Transformer (大模型)	28.4	41.8	$2.3 \cdot 10^{19}$	

残差 Dropout 我们对每个子层的输出应用 dropout [33]，再将其与子层输入相加并进行归一化。此外，我们还对编码器和解码器堆栈中嵌入表示与位置编码之和和应用 dropout。对于基础模型，我们采用 $P_{drop} = 0.1$ 的比率。

标签平滑 训练期间，我们采用了取值为 $\epsilon_{ls} = 0.1$ [36] 的标签平滑。这会损害困惑度，因为模型会变得更加不确定，但能提高准确率和 BLEU 分数。

6 结果

6.1 机器翻译

在 WMT 2014 英译德翻译任务中，大型 Transformer 模型（表 2 中的 Transformer (big)）比此前报告的最佳模型（包括集成模型）高出 2.0 个 BLEU 点，创下 28.4 的 BLEU 新纪录。该模型的配置列于表 3 的最后一行。使用 8 块 P100 GPU 训练耗时 3.5 天。即使是我们的基础模型，也以远低于任何有竞争力模型的训练成本，超越了此前发表的所有模型和集成模型。

在 WMT 2014 英译法翻译任务中，我们的大型模型取得了 41.0 的 BLEU 分数，超过了此前发表的所有单模型，而训练成本不到此前最先进模型的 1/4。英译法 Transformer (big) 模型的训练使用了 $P_{drop} = 0.1$ 的 dropout 率，而非 0.3。

对于基础模型，我们使用了一个通过对最后 5 个检查点取平均值而得到的单一模型，这些检查点每隔 10 分钟保存一次。对于大型模型，我们对最后 20 个检查点取平均值。我们使用了束宽为 4、长度惩罚系数为 $\alpha = 0.6$ [38] 的束搜索。这些超参数是在开发集上经过实验后选定的。推理时，我们将最大输出长度设为输入长度 + 50，但在可能的情况下提前终止 [38]。

表 2 总结了我们的结果，并将我们的翻译质量和训练成本与文献中其他模型架构进行了比较。我们通过将训练时间、使用的 GPU 数量，以及对每个 GPU 持续单精度浮点运算能力的估算值相乘，来估算训练模型所需的浮点运算次数 ⁵。

6.2 模型变体

为评估 Transformer 各个组件的重要性，我们以不同方式调整基础模型，并测量其在英语到德语翻译任务上的性能变化，实验在

⁵我们使用了 2.8、3.7、6.0 和 9.5 TFLOPS，分别对应 K80、K40、M40 和 P100，分别。

Table 3: Variations on the Transformer architecture. Unlisted values are identical to those of the base model. All metrics are on the English-to-German translation development set, newstest2013. Listed perplexities are per-wordpiece, according to our byte-pair encoding, and should not be compared to per-word perplexities.

	N	d_{model}	d_{ff}	h	d_k	d_v	P_{drop}	ϵ_{ls}	train steps	PPL (dev)	BLEU (dev)	params $\times 10^6$		
base	6	512	2048	8	64	64	0.1	0.1	100K	4.92	25.8	65		
(A)					1	512	512				5.29	24.9		
					4	128	128				5.00	25.5		
					16	32	32				4.91	25.8		
					32	16	16				5.01	25.4		
(B)					16					5.16	25.1	58		
					32					5.01	25.4	60		
(C)	2									6.11	23.7	36		
	4									5.19	25.3	50		
	8									4.88	25.5	80		
		256					32	32				5.75	24.5	28
		1024					128	128				4.66	26.0	168
			1024								5.12	25.4	53	
			4096								4.75	26.2	90	
(D)							0.0				5.77	24.6		
							0.2				4.95	25.5		
								0.0				4.67	25.3	
								0.2				5.47	25.7	
(E)	positional embedding instead of sinusoids									4.92	25.7			
big	6	1024	4096	16					0.3	300K	4.33	26.4	213	

development set, newstest2013. We used beam search as described in the previous section, but no checkpoint averaging. We present these results in Table 3.

In Table 3 rows (A), we vary the number of attention heads and the attention key and value dimensions, keeping the amount of computation constant, as described in Section 3.2.2. While single-head attention is 0.9 BLEU worse than the best setting, quality also drops off with too many heads.

In Table 3 rows (B), we observe that reducing the attention key size d_k hurts model quality. This suggests that determining compatibility is not easy and that a more sophisticated compatibility function than dot product may be beneficial. We further observe in rows (C) and (D) that, as expected, bigger models are better, and dropout is very helpful in avoiding over-fitting. In row (E) we replace our sinusoidal positional encoding with learned positional embeddings [9], and observe nearly identical results to the base model.

6.3 English Constituency Parsing

To evaluate if the Transformer can generalize to other tasks we performed experiments on English constituency parsing. This task presents specific challenges: the output is subject to strong structural constraints and is significantly longer than the input. Furthermore, RNN sequence-to-sequence models have not been able to attain state-of-the-art results in small-data regimes [37].

We trained a 4-layer transformer with $d_{\text{model}} = 1024$ on the Wall Street Journal (WSJ) portion of the Penn Treebank [25], about 40K training sentences. We also trained it in a semi-supervised setting, using the larger high-confidence and BerkleyParser corpora from with approximately 17M sentences [37]. We used a vocabulary of 16K tokens for the WSJ only setting and a vocabulary of 32K tokens for the semi-supervised setting.

We performed only a small number of experiments to select the dropout, both attention and residual (section 5.4), learning rates and beam size on the Section 22 development set, all other parameters remained unchanged from the English-to-German base translation model. During inference, we

表 3: Transformer 架构的变体。未列出的数值与基础模型相同。所有指标均在英译德翻译开发集 newstest2013 上计算。所列困惑度按词片段计算，依据我们使用的字节对编码，且不应与按词计算的困惑度进行比较。

	N	$d_{\text{模型}}$	d_{ff}	h	d_k	d_v	P_{drop}	ϵ_{ls}	训练 步数	PPL (开发集)	BLEU (开发集)	参数 $\times 10^6$	
base	6	512	2048	8	64	64	0.1	0.1	100K	4.92	25.8	65	
(A)					1	512	512				5.29	24.9	
					4	128	128				5.00	25.5	
					16	32	32				4.91	25.8	
					32	16	16				5.01	25.4	
(B)					16					5.16	25.1	58	
					32					5.01	25.4	60	
(C)	2									6.11	23.7	36	
	4									5.19	25.3	50	
	8									4.88	25.5	80	
		256				32	32			5.75	24.5	28	
		1024				128	128			4.66	26.0	168	
			1024							5.12	25.4	53	
			4096							4.75	26.2	90	
(D)								0.0		5.77	24.6		
								0.2		4.95	25.5		
								0.0		4.67	25.3		
								0.2		5.47	25.7		
(E)	使用位置嵌入而非正弦函数									4.92	25.7		
big	6	1024	4096	16				0.3	300K	4.33	26.4	213	

开发集 newstest2013 上进行。我们使用了上一节所述的束搜索，但没有进行检查点平均。结果如表 3 所示。

在表 3 的 (A) 行中，我们按照第 3.2.2 节所述，在保持计算量不变的情况下，改变注意力头的数量以及注意力键和值的维度。单头注意力的 BLEU 分数比最佳设置低 0.9 分，而注意力头过多也会导致质量下降。

在表 3 的 (B) 行中，我们观察到，减小注意力键的维度 d_k 会损害模型质量。这表明，确定兼容性并非易事，采用比点积更复杂的兼容性函数可能会有帮助。我们还从 (C) 行和 (D) 行观察到，正如预期的那样，更大的模型效果更好，而且 dropout 对避免过拟合非常有效。在 (E) 行中，我们用学习得到的位置嵌入 [9],替换了正弦位置编码，发现结果与基线模型几乎相同。

6.3 英语成分句法分析

为了评估 Transformer 能否泛化到其他任务，我们在英语成分句法分析任务上进行了实验。这项任务面临着一些特殊挑战：输出受到严格的结构约束，而且长度远大于输入。此外，RNN 序列到序列模型在小数据场景下尚未取得最先进的结果 [37]。

我们使用 $d_{\text{model}} = 1024$ 在宾夕法尼亚树库（Penn Treebank）的《华尔街日报》（WSJ）部分训练了一个 4 层 Transformer，该部分 [25], 约有 4 万个训练句子。我们还在半监督设置下训练了该模型，使用规模更大的高置信度语料库和 BerkleyParser 语料库，其中约有 1700 万个句子[37]。仅使用 WSJ 的设置采用了 1.6 万个词元的词汇表，半监督设置则采用了 3.2 万个词元的词汇表。

我们只进行了少量实验，以在第 22 节的开发集上选择 dropout 参数（包括注意力层和残差连接，见第 5.4 节）、学习率和束宽；其他所有参数均与英译德基础翻译模型保持一致。在推理过程中，我们

Table 4: The Transformer generalizes well to English constituency parsing (Results are on Section 23 of WSJ)

Parser	Training	WSJ 23 F1
Vinyals & Kaiser el al. (2014) [37]	WSJ only, discriminative	88.3
Petrov et al. (2006) [29]	WSJ only, discriminative	90.4
Zhu et al. (2013) [40]	WSJ only, discriminative	90.4
Dyer et al. (2016) [8]	WSJ only, discriminative	91.7
Transformer (4 layers)	WSJ only, discriminative	91.3
Zhu et al. (2013) [40]	semi-supervised	91.3
Huang & Harper (2009) [14]	semi-supervised	91.3
McClosky et al. (2006) [26]	semi-supervised	92.1
Vinyals & Kaiser el al. (2014) [37]	semi-supervised	92.1
Transformer (4 layers)	semi-supervised	92.7
Luong et al. (2015) [23]	multi-task	93.0
Dyer et al. (2016) [8]	generative	93.3

increased the maximum output length to input length + 300. We used a beam size of 21 and $\alpha = 0.3$ for both WSJ only and the semi-supervised setting.

Our results in Table 4 show that despite the lack of task-specific tuning our model performs surprisingly well, yielding better results than all previously reported models with the exception of the Recurrent Neural Network Grammar [8].

In contrast to RNN sequence-to-sequence models [37], the Transformer outperforms the Berkeley-Parser [29] even when training only on the WSJ training set of 40K sentences.

7 Conclusion

In this work, we presented the Transformer, the first sequence transduction model based entirely on attention, replacing the recurrent layers most commonly used in encoder-decoder architectures with multi-headed self-attention.

For translation tasks, the Transformer can be trained significantly faster than architectures based on recurrent or convolutional layers. On both WMT 2014 English-to-German and WMT 2014 English-to-French translation tasks, we achieve a new state of the art. In the former task our best model outperforms even all previously reported ensembles.

We are excited about the future of attention-based models and plan to apply them to other tasks. We plan to extend the Transformer to problems involving input and output modalities other than text and to investigate local, restricted attention mechanisms to efficiently handle large inputs and outputs such as images, audio and video. Making generation less sequential is another research goals of ours.

The code we used to train and evaluate our models is available at <https://github.com/tensorflow/tensor2tensor>.

Acknowledgements We are grateful to Nal Kalchbrenner and Stephan Gouws for their fruitful comments, corrections and inspiration.

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表 4: Transformer 在英语成分句法分析任务上具有良好的泛化能力（结果基于 WSJ 第 23 节）

解析器	训练	WSJ 第 23 节 F1
Vinyals 和 Kaiser 等人 (2014) [37]	仅使用 WSJ, 判别式	88.3
Petrov 等 (2006) [29]	仅使用 WSJ, 判别式	90.4
Zhu 等 (2013) [40]	仅使用 WSJ, 判别式	90.4
Dyer 等 (2016) [8]	仅使用 WSJ, 判别式	91.7
Transformer (4层)	仅限 WSJ, 判别式	91.3
Zhu 等 (2013) [40]	半监督	91.3
Huang 和 Harper (2009) [14]	半监督	91.3
McClosky 等 (2006) [26]	半监督	92.1
Vinyals 和 Kaiser 等 (2014) [37]	半监督	92.1
Transformer (4 层)	半监督	92.7
Luong 等人 (2015) [23]	多任务	93.0
Dyer 等人 (2016) [8]	生成式	93.3

incr将最大输出长度设为输入长度 + 300。我们使用了 21 的束宽，并且 $\alpha = 0.3$ for 无论是在仅使用 WSJ 的设置中，还是在半监督设置中。

表 4 中的结果表明，尽管没有针对特定任务进行调优，我们的模型仍表现出色，取得了优于此前所有已报告模型的结果，唯一例外是循环神经网络语法 [8]。

与 RNN 序列到序列模型 [37],相比，即使仅使用包含 4 万个句子的 WSJ 训练集进行训练，Transformer 的表现也优于 Berkeley 句法分析器 [29]。

7 结论

在这项工作中，我们提出了 Transformer，这是第一个完全基于注意力机制的序列转换模型。它以多头自注意力机制取代了编码器-解码器架构中最常用的循环层。

对于翻译任务，Transformer 的训练速度显著快于基于循环层或卷积层的架构。在 WMT 2014 英译德和 WMT 2014 英译法两项翻译任务中，我们都取得了新的最先进结果。在前一项任务中，我们表现最好的模型甚至优于此前报告的所有集成模型。

我们对基于注意力机制的模型的未来充满期待，并计划将其应用于其他任务。我们计划将 Transformer 扩展到涉及文本以外输入和输出模态的问题，并研究局部、受限的注意力机制，以高效处理图像、音频和视频等大型输入和输出。让生成过程减少对序列的依赖，也是我们的另一项研究目标。

用于训练和评估我们模型的代码可在 <https://github.com/tensorflow/tensor2tensor> 获取。

致谢 我们感谢 Nal Kalchbrenner 和 Stephan Gouws 提出的富有成效的 co意见、修正和启发。

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Attention Visualizations

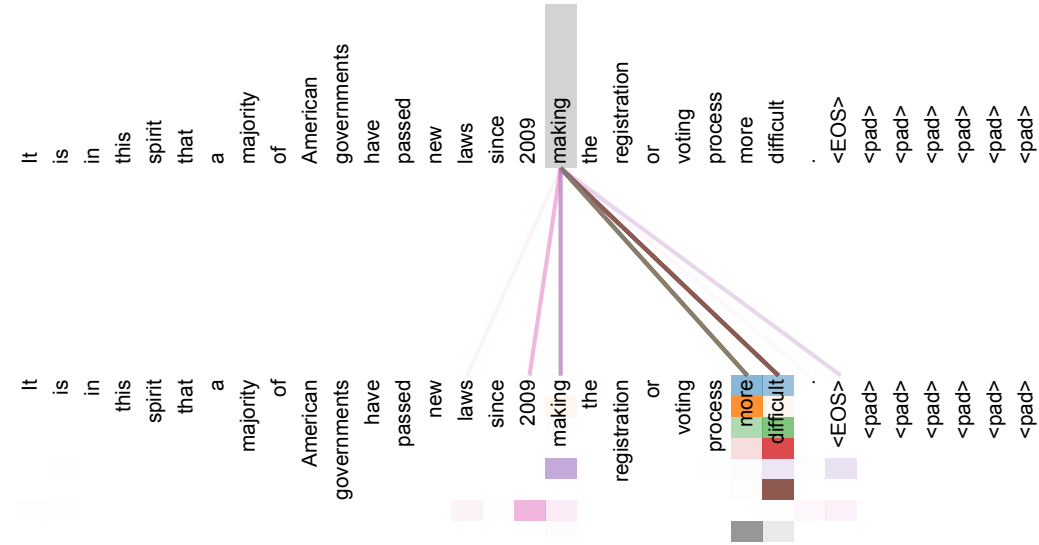


Figure 3: An example of the attention mechanism following long-distance dependencies in the encoder self-attention in layer 5 of 6. Many of the attention heads attend to a distant dependency of the verb ‘making’, completing the phrase ‘making...more difficult’. Attentions here shown only for the word ‘making’. Different colors represent different heads. Best viewed in color.

注意力可视化

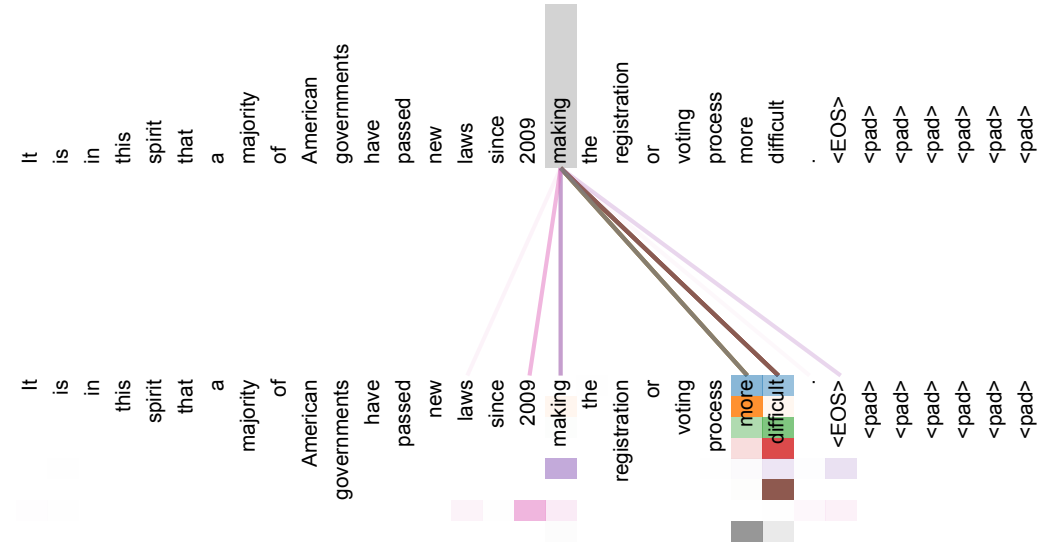


图 3: 编码器第 5/6 层自注意力中, 注意力机制追踪长距离依赖关系的示例。许多注意力头都关注动词 “making” 的远距离依赖关系, 从而补全短语 “making...more difficult”。此处仅显示针对单词 “making” 的注意力。不同颜色代表不同的注意力头。彩色查看效果最佳。

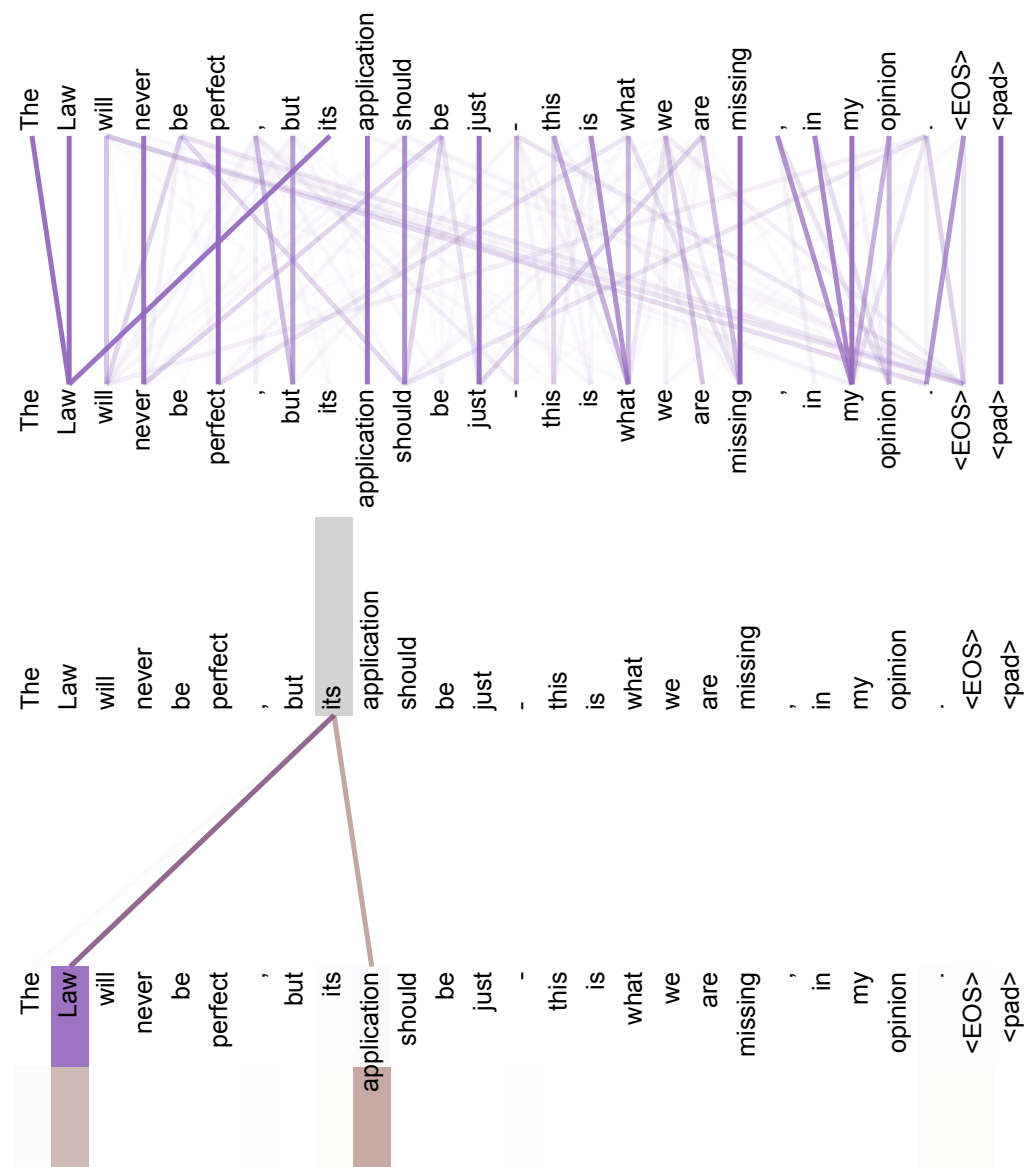


Figure 4: Two attention heads, also in layer 5 of 6, apparently involved in anaphora resolution. Top: Full attentions for head 5. Bottom: Isolated attentions from just the word ‘its’ for attention heads 5 and 6. Note that the attentions are very sharp for this word.

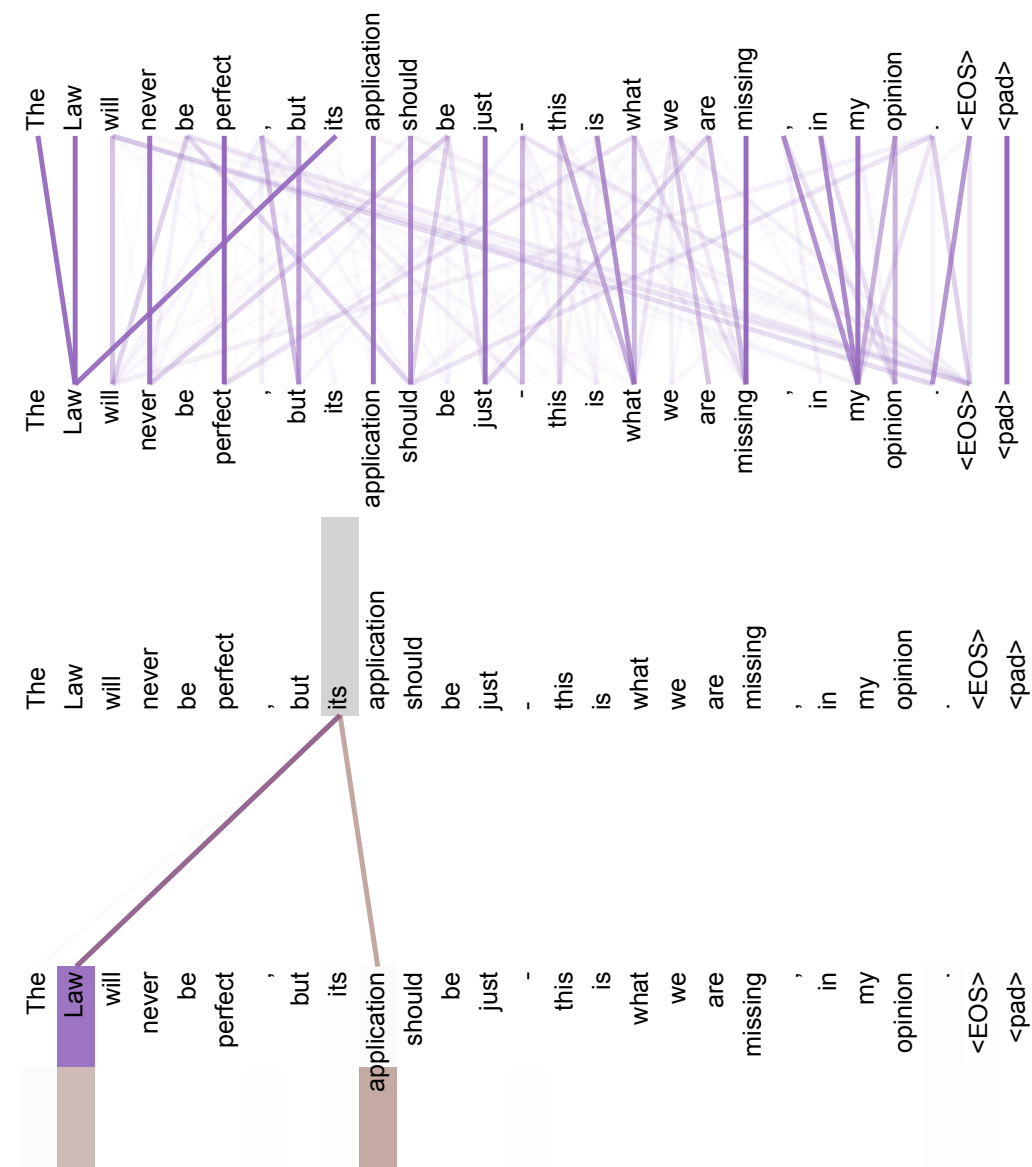


图4：第5层（共6层）中的两个注意力头似乎也参与了解决照应问题。上图：注意力头5的完整注意力分布。下图：仅显示注意力头5和6对单词“its”的注意力分布。请注意，该单词对应的注意力分布非常尖锐。

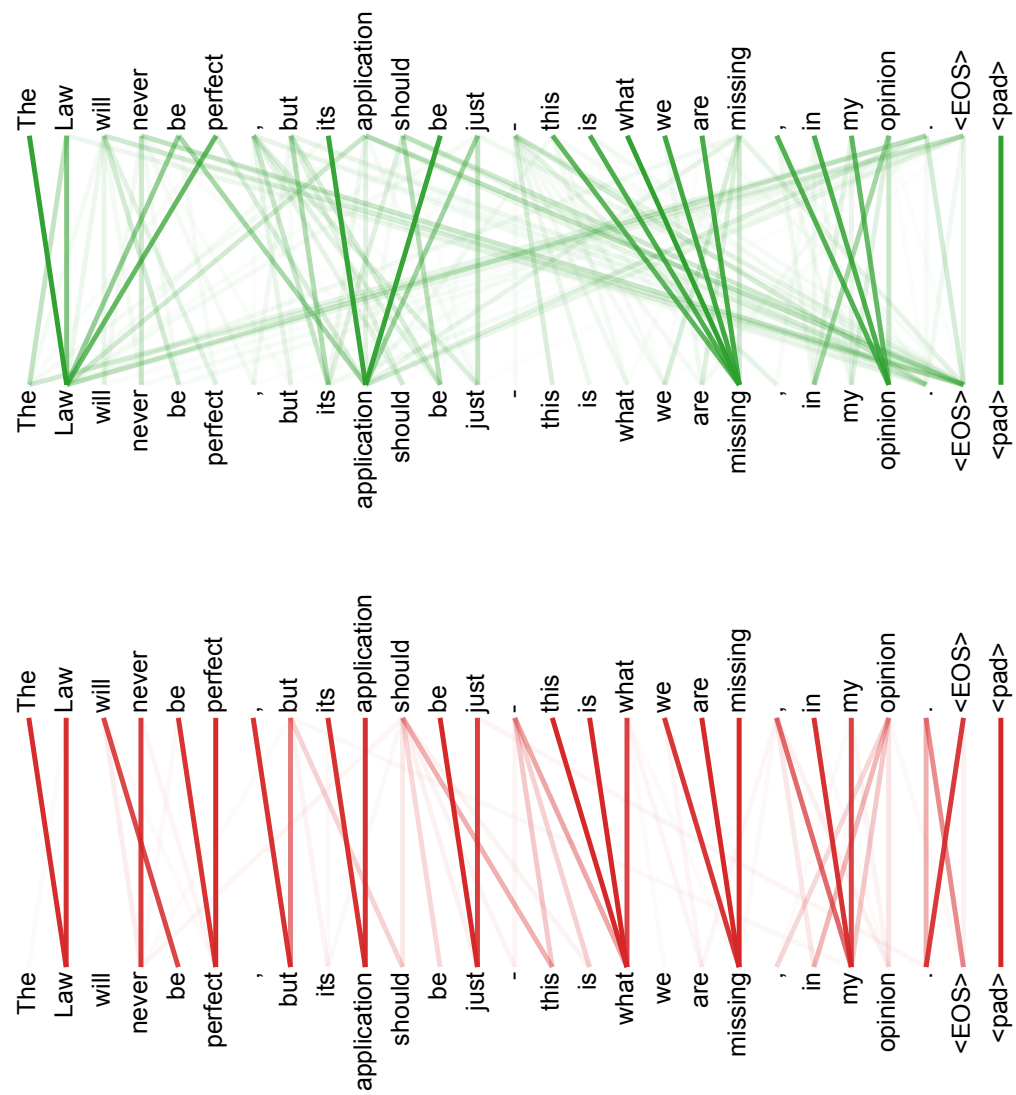


Figure 5: Many of the attention heads exhibit behaviour that seems related to the structure of the sentence. We give two such examples above, from two different heads from the encoder self-attention at layer 5 of 6. The heads clearly learned to perform different tasks.

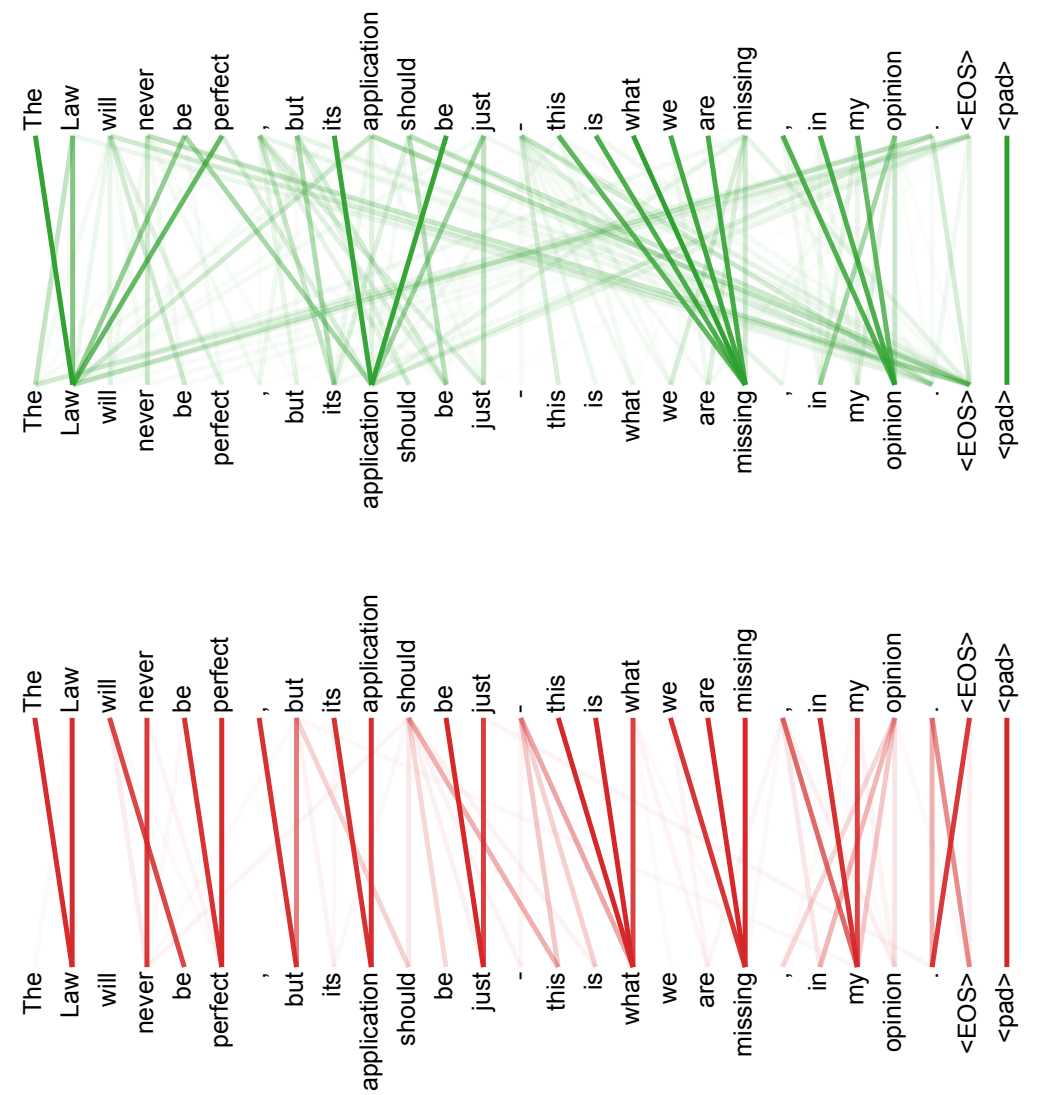


图5: 许多注意力头表现出的行为似乎与句子结构有关。上面给出了两个这样的例子, 分别来自编码器第6层中的第5层的两个不同自注意力头。这些注意力头显然学会了执行不同的任务。